100 Days of Linux - Day 045

Title: Viewing Running Processes with ps

Today's Goal
Learn how to view running processes using the **ps** command and understand the information it displays.

Why This Matters
Every application running on Linux is a process. Developers and data engineers use **ps** to troubleshoot programs, monitor scripts, and identify background jobs.

Command of the Day: ps

Common Syntax:
ps
ps -e (show all processes)
ps -ef (full-format list)
ps aux (detailed process list)

Important Columns:
PID = Process ID
TTY = Terminal
TIME = CPU time used
CMD = Command that started the process

Beginner Lesson
A process is simply a program that is currently running. The **ps** command lets you inspect these processes. Every process has a unique Process ID (PID), which you'll use in later lessons when learning how to stop processes.

Practice Questions

1. Run the ps command and observe the processes attached to your current terminal.
2. Run ps -e and count how many processes are displayed.
3. Run ps -ef and identify the PID of your shell (bash or zsh if shown).
4. Run ps aux and locate the process running the WSL terminal or your current shell.
5. Display your current working directory after completing today's tasks.

Hints

Compare the outputs of ps, ps -e, ps -ef, and ps aux to understand how much additional information each option provides.

Mini Challenge

Open a second terminal window. Run a command like 'sleep 300' in one terminal, then use ps in the other terminal to locate the sleep process and note its PID.

Common Mistakes

Confusing the PID with the PPID (Parent Process ID). They are different values with different purposes.

Did You Know?

Every process on Linux ultimately traces back to a parent process, creating a process tree.

Pro Tip

The PID is one of the most important values in Linux because many management commands use it.

Answer Key (Instructor Only)

Q1: ps
Q2: ps -e
Q3: ps -ef
Q4: ps aux
Q5: pwd

Homework

Run ps at different times during the day and observe how the process list changes as you open and close applications.

Next Day Preview

Tomorrow you'll learn the top command to monitor CPU, memory, and running processes in real time.